

In fact, two different viruses were observed performing in this way. The first virus was spread by an MS-DOS setup diskette while the second virus was transmitted using a still undiscovered method. While we observed no boot sector infections of PC based Unix systems during 1994, we received reports from system administrators who were requesting information on our Unix anti-virus product because they had experienced hundreds of infections during 1995. In one instance, a single multinational company lost its entire international network overnight. The estimated cost in lost time, resources, and sales was in the millions of dollars.

Once it is understood that the BIOS and processor functions are the same for both operating systems, it is very easy to see how a trans-platform virus could be designed by intention. The virus would be able to process correctly by inspecting the operating system using only common BIOS calls and then modify its basic behaviour using a simple “if” structure.

Traditional Categories Of Protection And Their Failure

There are three traditional categories of protection, none of which provide complete or significant protection as stand-alone methods of implementation. The categories are Control, Inspection and Integrity. Each of these methods has traditionally been used separately.

Control has been the primary intent of the U.S. national standards on computer security. They deal with the control of access to the system, its functions, resources and the ability to move or share data in the system. These national standards are codified in a library generally referred to as the Rainbow series. (The name was given because the books have different colour covers making a library shelf look like a rainbow.) While these standards are a valuable and important aspect of computer security, they do not provide a deterrent against software attack. A virus is an effective way of gaining control over a system, even a highly controlled system such as a B1 rated version of Unix. In this case, control does not provide protection